

EXPERIMENT 8

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Experiment 8 -- To study and implement Views

Aim : To study and learn the about the Views, grant or revoke submission.

Question :8 Part A (i): Complex View

SQL Query:

```
CREATE TABLE customer_master
```

```
(  
    customer_id VARCHAR(5) PRIMARY KEY,  
    full_name VARCHAR(50) NOT NULL,    city  
    VARCHAR(30)  
);
```

```
CREATE TABLE product_catalog
```

```
(  
    product_id VARCHAR(5) PRIMARY KEY,  
    product_name VARCHAR(50) NOT NULL,  
    unit_price NUMERIC(10,2) NOT NULL  
);
```

```
CREATE TABLE sales_orders
```

EXPERIMENT 8

```
(  
  
    order_id SERIAL PRIMARY KEY,  
    product_id VARCHAR(5)  
        REFERENCES product_catalog(product_id),  
  
    quantity INT NOT NULL,  
  
    customer_id VARCHAR(5)  
        REFERENCES customer_master(customer_id),  
  
    discount_percent NUMERIC(5,2),  
  
    order_date DATE NOT NULL  
);
```

```
INSERT INTO customer_master  
(customer_id, full_name, city)  
VALUES  
('C1', 'Amit Sharma', 'Delhi'),  
('C2', 'Priya Verma', 'Mumbai'),  
('C3', 'Ravi Kumar', 'Bangalore'),  
('C4', 'Neha Singh', 'Kolkata'),  
('C5', 'Arjun Mehta', 'Hyderabad');
```

```
INSERT INTO product_catalog  
(product_id, product_name, unit_price)  
VALUES  
('P1', 'Smartphone X100', 25000),
```

EXPERIMENT 8

('P2', 'Laptop Pro 15', 65000),
('P3', 'Wireless Earbuds', 5000),

('P4', 'Smartwatch Fit', 30000),
('P5', 'Gaming Console', 45000);

```
INSERT INTO sales_orders
(product_id, quantity, customer_id, discount_percent, order_date)
VALUES
('P1', 2, 'C1', 5, '2025-09-01'),
('P2', 1, 'C2', 10, '2025-09-02'),
('P3', 3, 'C3', 0, '2025-09-03'),
('P4', 1, 'C4', 8, '2025-09-04'),
('P5', 1, 'C5', 12, '2025-09-05'),
('P1', 1, 'C2', 5, '2025-09-06'),
('P3', 2, 'C1', 0, '2025-09-07');
```

```
CREATE OR REPLACE VIEW vw_sales_report as
select t.order_id, cc.full_name, t.quantity, t.final_am as final_amount
FROM (
    select so.discount_percent, so.product_id, so.customer_id, so.order_id, so.quantity, (
        (pc.unit_price * so.quantity * so.discount_percent )/100
    ) as final_am
FROM sales_orders as so
JOIN product_catalog as pc
ON so.product_id = pc.product_id
```

EXPERIMENT 8

) t

JOIN customer_master as cc

ON cc.customer_id =t.customer_id

SELECT * from vw_sales_report

CREATE USER Exp_8_user with PASSWORD '1234';

GRANT SELECT ON vw_sales_report TO Exp_8_user;

Question: Part A (ii) Materialized view for Category_report_generation SQL
Query:

DROP TABLE IF EXISTS category_master CASCADE;

DROP TABLE IF EXISTS product_master CASCADE;

DROP TABLE IF EXISTS customer_master CASCADE;

DROP TABLE IF EXISTS order_details CASCADE;

```
CREATE TABLE category_master (  
category_id VARCHAR(5) PRIMARY KEY,  
category_name VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE product_master (  
product_id VARCHAR(5) PRIMARY KEY,  
product_name VARCHAR(50) NOT NULL,  
category_id VARCHAR(5)  
REFERENCES category_master(category_id)
```

EXPERIMENT 8

);

CREATE TABLE customer_master (

customer_id VARCHAR(5) PRIMARY KEY,
customer_name VARCHAR(50) NOT NULL
);

CREATE TABLE order_details (

order_id VARCHAR(5) NOT NULL,
customer_id VARCHAR(5)
REFERENCES customer_master(customer_id),
product_id VARCHAR(5)
REFERENCES product_master(product_id),
quantity INT NOT NULL, unit_price
NUMERIC(10,2) NOT NULL
);

INSERT INTO category_master

(category_id, category_name)

VALUES

('C1', 'Electronics'),

('C2', 'Clothing'),

('C3', 'Books');

INSERT INTO product_master

(product_id, product_name, category_id)

EXPERIMENT 8

VALUES

('P101', 'Laptop', 'C1'),

('P102', 'Headphones', 'C1'),

('P201', 'Jacket', 'C2'),

('P202', 'Shoes', 'C2'),

('P301', 'SQL Book', 'C3');

INSERT INTO customer_master

(customer_id, customer_name)

VALUES

('U1', 'Aman'),

('U2', 'Priya'),

('U3', 'Rahul'),

('U4', 'Simran');

INSERT INTO order_details

(order_id, customer_id, product_id, quantity, unit_price)

VALUES

('O101', 'U1', 'P101', 1, 5000),

('O102', 'U2', 'P102', 2, 1000),

('O103', 'U1', 'P201', 1, 3000),

('O104', 'U3', 'P202', 2, 2000),

('O105', 'U4', 'P301', 3, 500),

('O106', 'U2', 'P101', 1, 5000),

('O107', 'U3', 'P201', 1, 3000),

('O108', 'U4', 'P102', 1, 1000);

EXPERIMENT 8

DROP MATERIALIZED VIEW mw_category_report

EXPLAIN ANALYZE

CREATE MATERIALIZED VIEW mw_category_report

AS

```
select    cm.category_id    as
CatID,

          cm.category_name as CatName,

          SUM(od.quantity) as TOTAL_PRODUCT,

          COUNT(DISTINCT od.order_id) as TOTAL_ORDERS,

          COUNT(DISTINCT od.customer_id) as Unique_Customer,

          SUM(od.quantity * od.unit_price) as TOTAL_REVENUE,

          ROUND( SUM(od.quantity * od.unit_price)/COUNT(DISTINCT od.order_id) ,2) as AVG_REVENUE,

          (CASE WHEN SUM(od.quantity * od.unit_price) >= 10000 THEN 'Excellent'

          WHEN SUM(od.quantity * od.unit_price) >= 6000 THEN 'Good'

          WHEN SUM(od.quantity * od.unit_price) >= 3000 THEN 'Average'  ELSE 'Low' END)

as Category_performance

from Order_details as od

JOIN product_master as pm

ON od.product_id = pm.product_id

JOIN category_master as cm

ON cm.category_id = pm.category_id

GROUP BY cm.category_id, cm.category_name

SELECT * FROM mw_category_report
```

Learning Outcome

EXPERIMENT 8

After completing this experiment, the learner will be able to:

1. **Understand and implement Views**
2. **Apply aggregate functions** such as COUNT() and DISTINCT to get unique customer;
3. **Use joins and filtering techniques** to combine data from multiple tables and generate meaningful reports.